

Foundations of Prompt engineering

[Prompt engineering]

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Context engineering Context engineering is a related process that focuses on the context elements that accompany user prompts, which include system instructions, retrieved knowledge, tool definitions, conversation summaries, and task metadata. Context engineering is performed to improve reliability, provenance and token efficiency in production LLM systems. The concept emphasises operational practices such as token budgeting, provenance tags, versioning of context artifacts, observability (logging which context was supplied), and context regression tests to ensure that changes to supplied context do not silently alter system behaviour.

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GraphRAG (coined by Microsoft Research) is a technique that extends RAG with the use of a knowledge graph to allow the model to connect disparate pieces of information, synthesize insights, and understand summarized semantic concepts over large data collections. It was shown to be effective on datasets like the Violent Incident Information from News Articles.

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Automated prompt generation Recent research has explored automated prompt engineering, using optimization algorithms to generate or refine prompts without human intervention. These automated approaches aim to identify effective prompt patterns by analyzing model gradients, reinforcement feedback, or evolutionary processes, reducing the need for manual experimentation.

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Prompt A prompt is some natural language text that describes and prescribes the task that an artificial intelligence (AI) should perform. A prompt for a text-to-text language model can be a query, a command, or a longer statement referencing context, instructions, and conversation history. The process of prompt engineering may involve designing clear queries, refining wording, providing relevant context, specifying the style of output, and assigning a character for the AI to mimic in order to guide the model toward more accurate, useful, and consistent responses. When communicating with a text-to-image or a text-to-audio model, a typical prompt contains a description of a desired output such as "a high-quality photo of an astronaut riding a horse" or "Lo-fi slow BPM electro chill with organic samples". Prompt engineering may be applied to text-to-image models to achieve a desired subject, style, layout, lighting, and aesthetic.

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MIPRO (Multi-prompt Instruction Proposal Optimizer) optimizes the instructions and few-shot demonstrations of multi-stage language model programs, proposing small changes to module prompts and retaining those that improve a downstream performance metric without access to module-level labels or gradients. **GEPA** (Genetic-Pareto) is a reflective prompt optimizer for compound AI systems that combines language-model-based analysis of execution traces and textual feedback with a Pareto-based evolutionary search over a population of candidate systems; across four tasks, GEPA reports average gains of about 10% over reinforcement-learning-based Group Relative Policy Optimization (GRPO) and over 10% over the MIPROv2 prompt optimizer, while using up to 35 times fewer rollouts than GRPO. Open-source frameworks such as DSPy and Opik expose these and related optimizers, allowing prompt search to be expressed as part of a programmatic pipeline rather than through manual trial and error.

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Using language models to generate prompts LLMs themselves can be used to compose prompts for LLMs. The automatic prompt engineer algorithm uses one LLM to beam search over prompts for another LLM:

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Multi-shot A prompt may include a few examples for a model to learn from in context, an approach called few-shot learning. For example, the prompt may ask the model to complete "maison → house, chat → cat, chien →", with the expected response being dog.

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History Early precedents of structured user interaction with ruled-based AI systems can be found in enterprise automation software from 1990s. For example, The Intelligent Filing Manager (1999), developed by Krishna C. Mukherjee, used a dynamic Q&A interface driven by rule-based expert system to collect user inputs for generating regulatory filings automatically across jurisdictions. While not involving neural networks, such systems featured prompt-like workflows that influenced later human-in-the-loop AI designs. In 2018, researchers first proposed that all previously separate tasks in natural language processing (NLP) could be cast as question-answer problems over a context. In addition, they trained a first single, joint, multi-task model that would answer any task-related question like "What is the sentiment" or "Translate this sentence to German" or "Who is the president?" The AI boom saw an increased focus within academic literature and professional practice on applying prompting techniques to get the model to output the desired outcome and avoid nonsensical output, a process characterized by trial-and-error. After the release of ChatGPT in 2022, prompt engineering was soon seen as an important business skill; companies began hiring dedicated prompt engineers, although, given advances in AI's ability to generate prompts better than humans, the employment market for prompt engineers has faced uncertainty. According to The Wall Street Journal in 2025, the job of prompt engineer was one of the hottest in 2023, but has become obsolete due to models that better intuit user intent and to company trainings. A repository for prompts reported that over 2,000 public prompts for around 170 datasets were available in February 2022. In 2022, the chain-of-thought prompting technique was proposed by Google researchers. In 2023, several text-to-text and text-to-image prompt databases were made publicly available. The Personalized Image-Prompt (PIP) dataset, a generated image-text dataset that has been categorized by 3,115 users, has also been made available publicly in 2024.

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Tree-of-thought Tree-of-thought prompting generalizes chain-of-thought by generating multiple lines of reasoning in parallel, with the ability to backtrack or explore other paths. It can use tree search algorithms like breadth-first, depth-first, or beam.

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Retrieval-augmented generation (RAG) Retrieval-augmented generation is a technique that enables GenAI models to retrieve and incorporate new information. It modifies interactions with an LLM so that the model responds to user queries with reference to a specified set of documents, using this information to supplement information from its pre-existing training data. This allows LLMs to use domain-specific and/or updated information. RAG improves large language models by incorporating information retrieval before generating responses. Unlike traditional LLMs that rely on static training data, RAG pulls relevant text from databases, uploaded documents, or web sources. By dynamically retrieving information, RAG enables AI to generate more accurate responses and fewer AI hallucinations without frequent retraining.

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Prompting to estimate model sensitivity Research consistently demonstrates that LLMs are highly sensitive to subtle variations in prompt formatting, structure, and linguistic properties. Some studies have shown up to 76 accuracy points across formatting changes in few-shot settings. Linguistic features significantly influence prompt effectiveness—such as morphology, syntax, and lexico-semantic changes—which meaningfully enhance task performance across a variety of tasks. Clausal syntax, for example, improves consistency and reduces uncertainty in knowledge retrieval. This sensitivity persists even with larger model sizes, additional few-shot examples, or instruction tuning. To address sensitivity of models and make them more robust, several evaluative methods have been proposed. FormatSpread facilitates systematic analysis by evaluating a range of plausible prompt formats, offering a more comprehensive performance interval. Similarly, PromptEval estimates performance distributions across diverse prompts, enabling robust metrics such as performance quantiles and accurate evaluations under constrained budgets.